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Paper Title:

**State-of-Charge Estimation of Li-Ion Battery in
Electric Vehicles Using Deep Neural Network**



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INTRODUCTION & BRIEF REVIEW



- The adoption of electric vehicles (EVs) is a key solution for reducing carbon emissions and addressing the global energy crisis.
- The efficiency and dependability of EVs largely depend on their energy storage systems, especially lithium-ion batteries (LIBs).
- LIBs are intricate and dynamic systems, influenced by multiple interconnected variables and nonlinear behaviors.
- A sophisticated Battery Management System (BMS) is essential to ensure the safe and optimal operation of LIBs.
- One of the BMS critical functions is monitoring the State of Charge (SoC), which represents the battery's current energy level.



INTRODUCTION & BRIEF REVIEW



- SoC represents the remaining charge as a percentage of the battery's rated capacity.
- Mathematically, SoC can be expressed as:

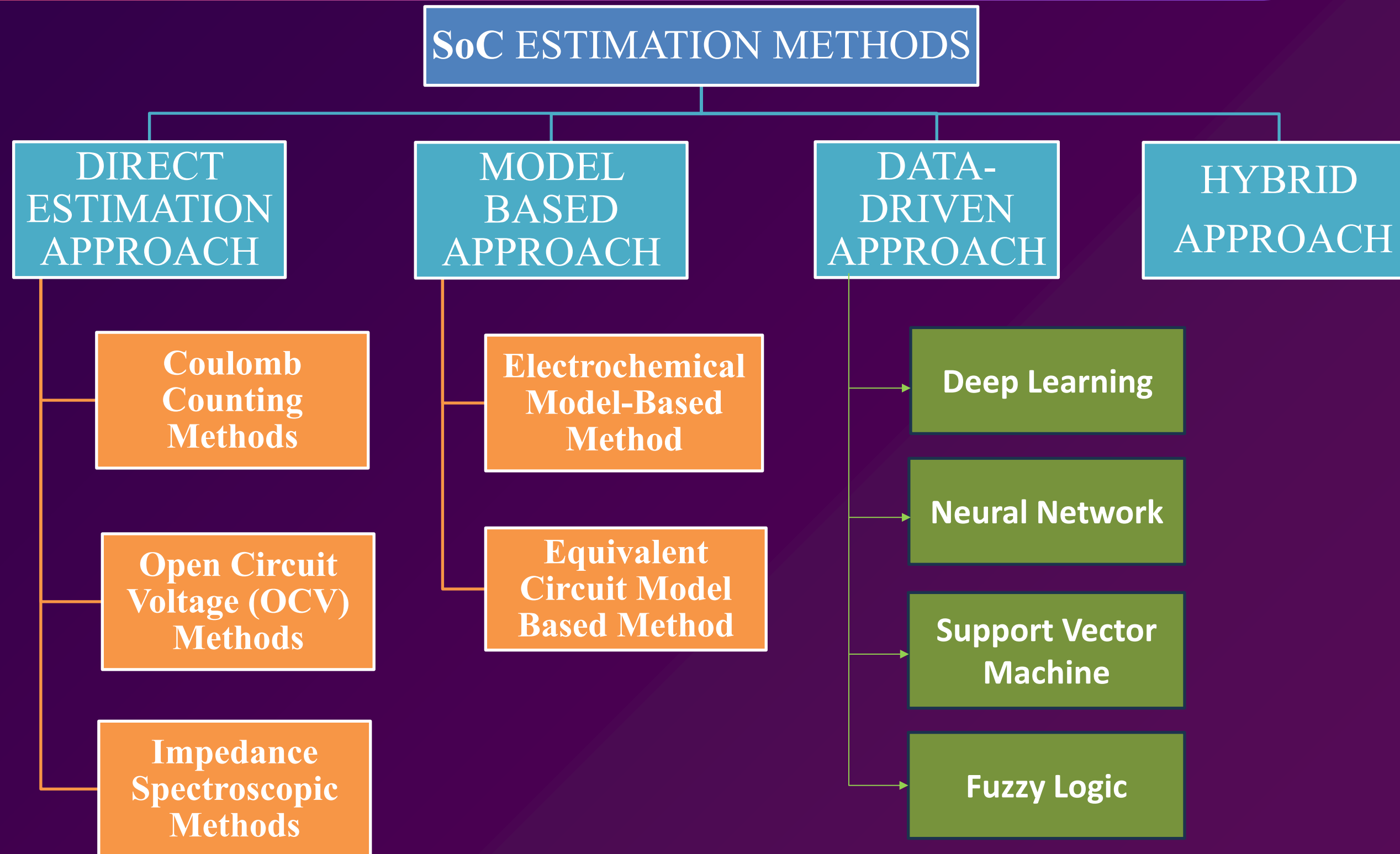
$$SoC = \frac{Q(t)}{Q_{rated}}$$

Q represents the remaining battery charge at time t and, Q_{rated} the rated battery capacity

- However, due to the nonlinearity, time-varying characteristics, and highly sophisticated battery dynamics, direct measurement of SoC is not feasible.
- Moreover, battery performance is substantially affected by aging, temperature variation and charge-discharge cycles.
- Consequently, determining precise battery SoC remains a critical yet challenging task.

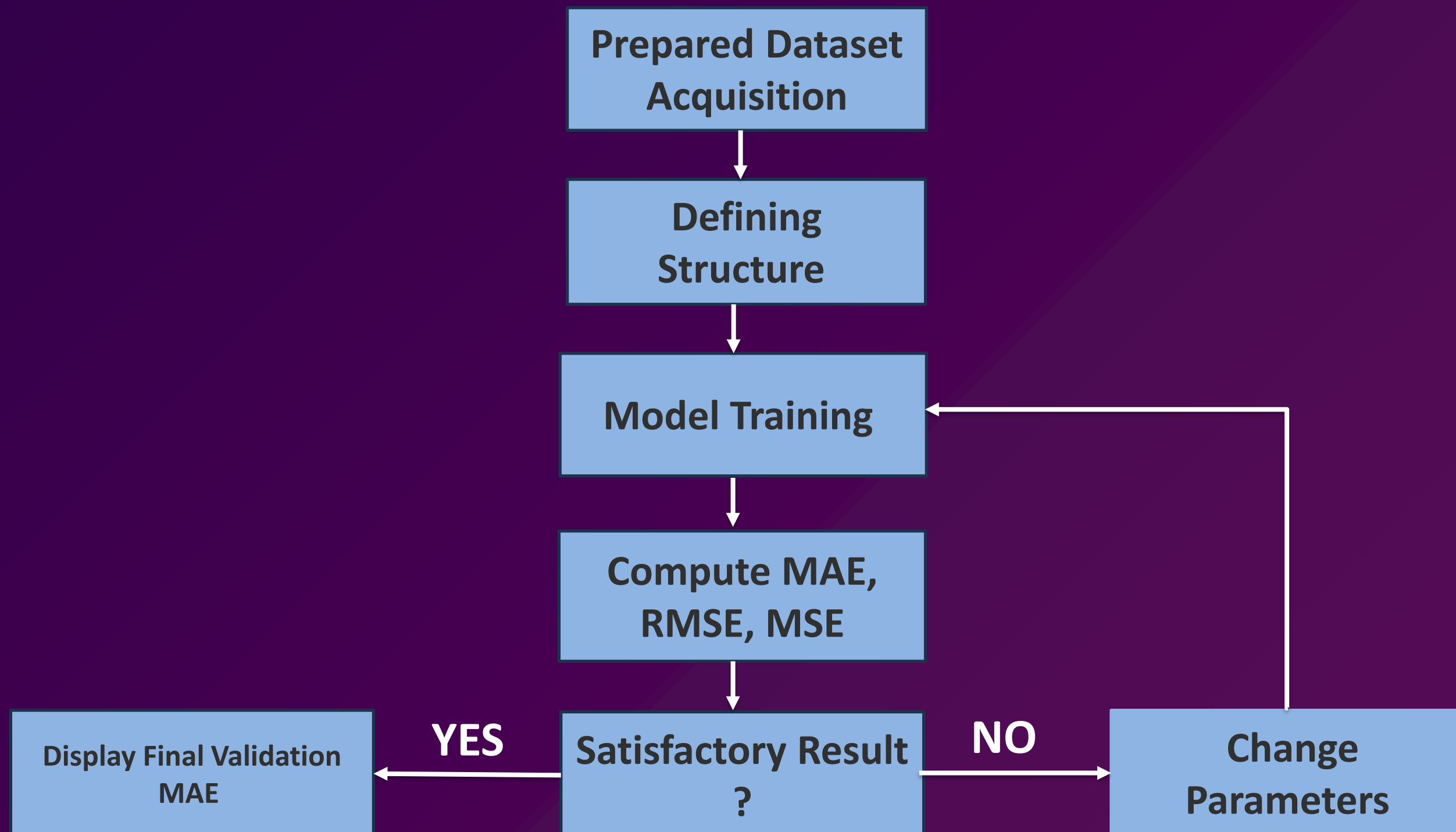


SoC ESTIMATION METHODS





METHODOLOGY





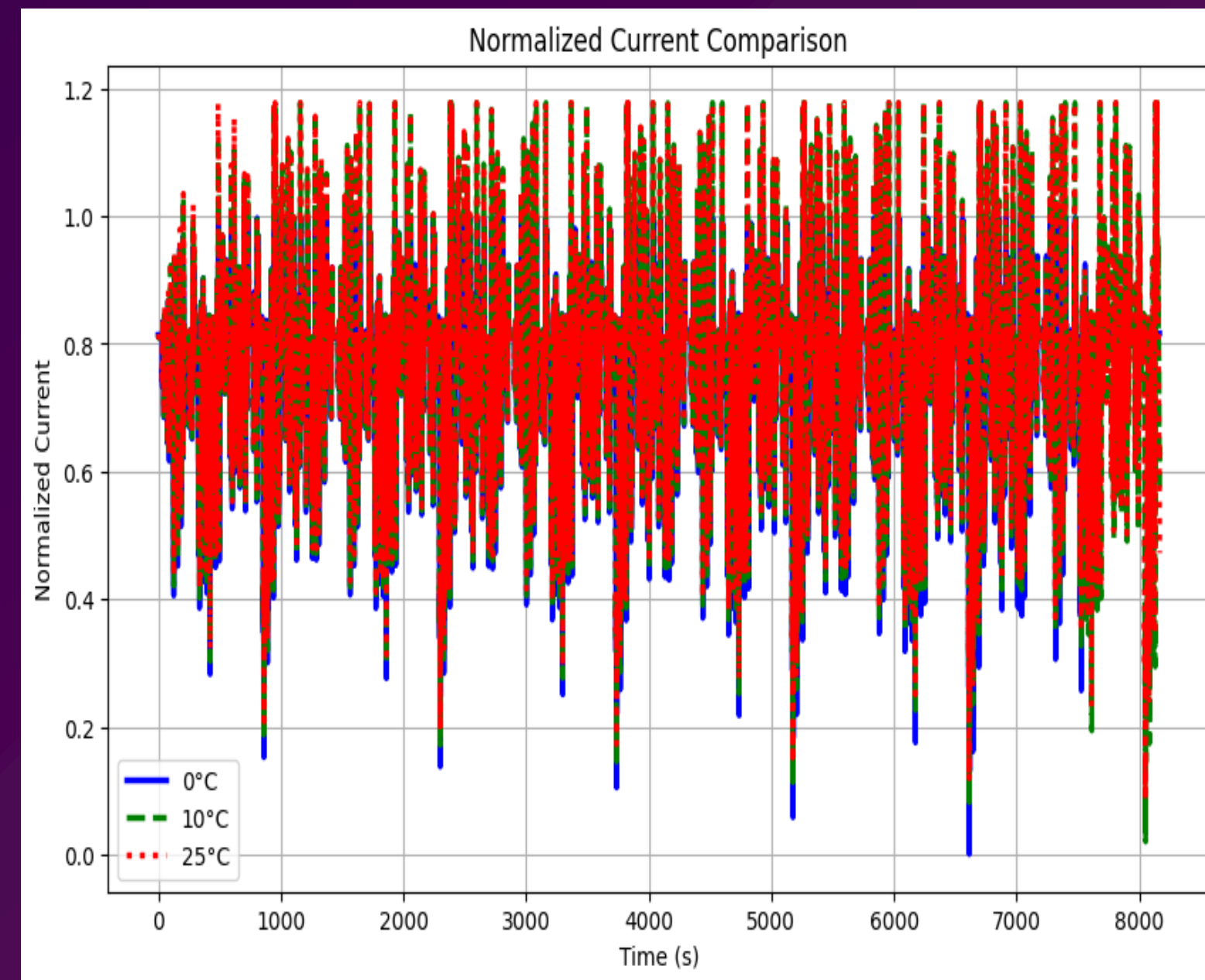
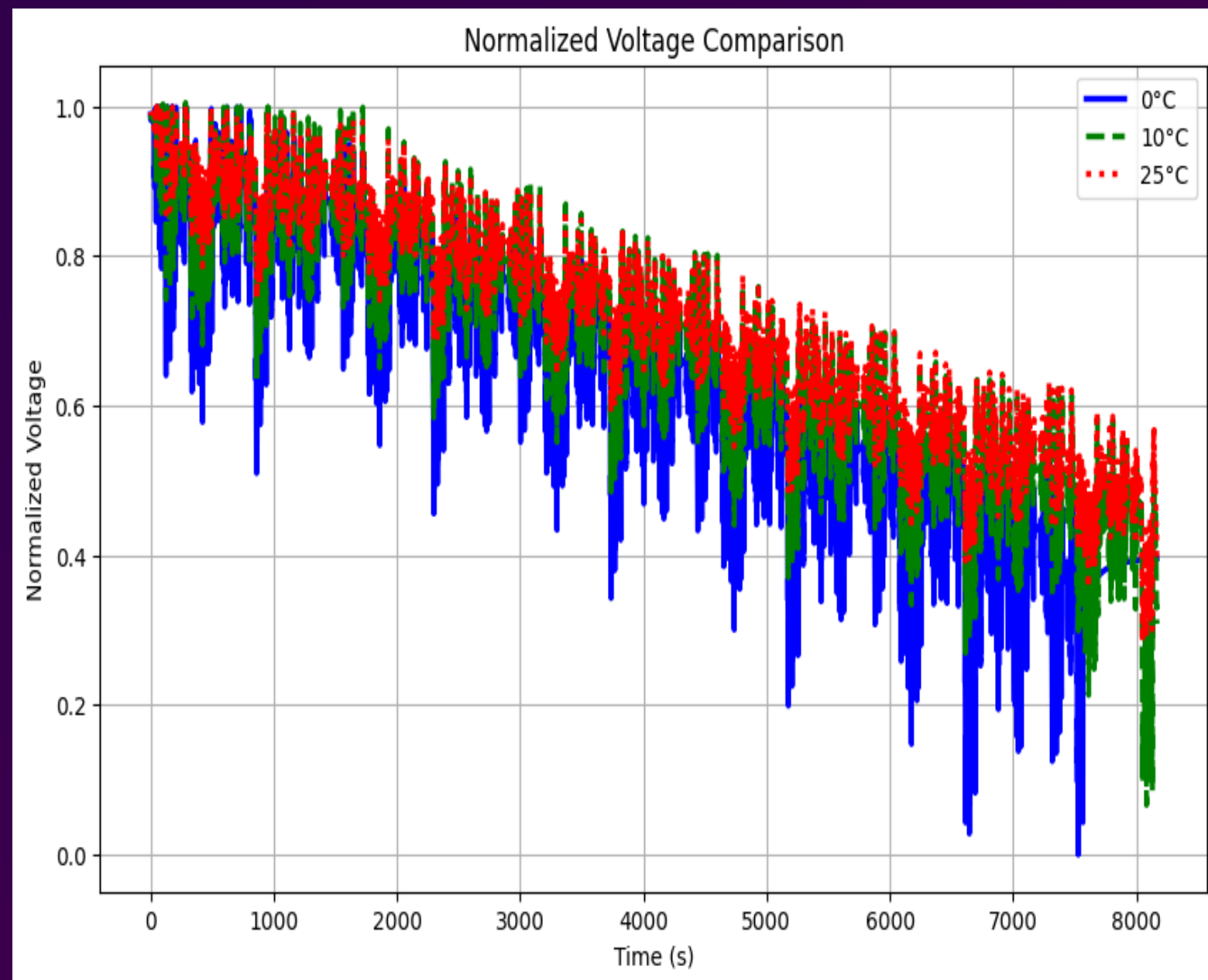
SoC ESTIMATION USING DATA-DRIVEN METHOD



- The reason for proposing DNN for SOC estimation is its effectiveness in simulating the non-linear behavior of the system.
- DNN is a structural connection of various neurons comprising input layers, output layers, and hidden layers.
- The proposed DNN consists of two hidden layers, with each hidden layer having 64 neurons.
- The input layer consists of three neurons, taking voltage, current, and temperature as inputs from the battery dataset.
- The output layer provides the instantaneous State of Charge (SOC).
- Iterations are set to minimize the cost function, which measures how close the predicted values are to the actual value.

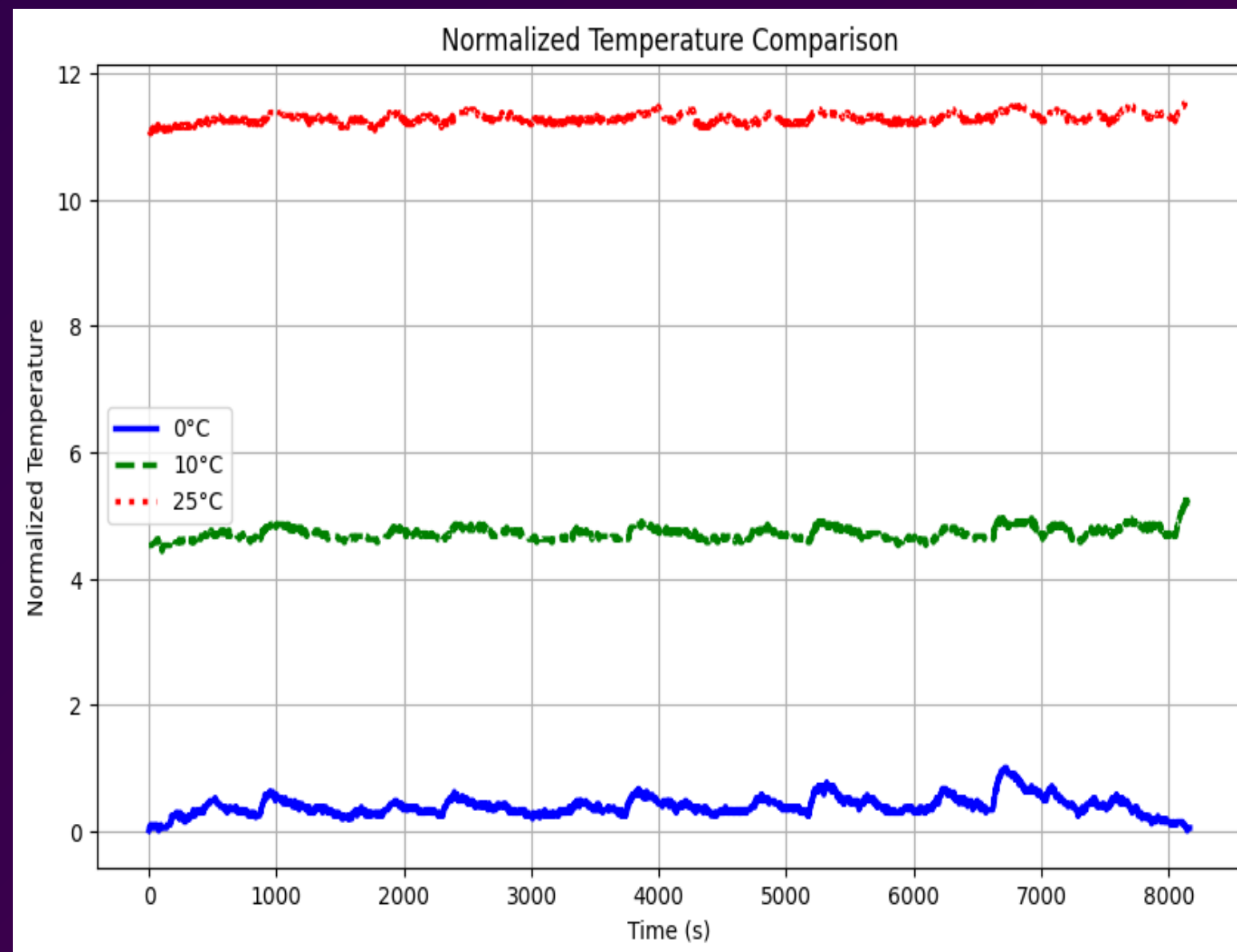


INPUT TO THE DATA-DRIVEN MODELS





INPUT TO THE DATA-DRIVEN MODELS



Battery type	LG18650HG2
Nominal capacity	3 Ampere-hour
Nominal voltage	3.6 Volt
Nominal current	3 Ampere
Maximum voltage	4.2 Volt
Cut-off voltage	2.5 Volt



SoC ESTIMATION USING DEEP NEURAL NETWORK (DNN)

The input and output vectors are expressed as:

$$X = [V, T, I]$$

$$Y = \text{SoC}$$

Let w and b represent the weights and bias respectively, then activation h of hidden layer is calculated as:

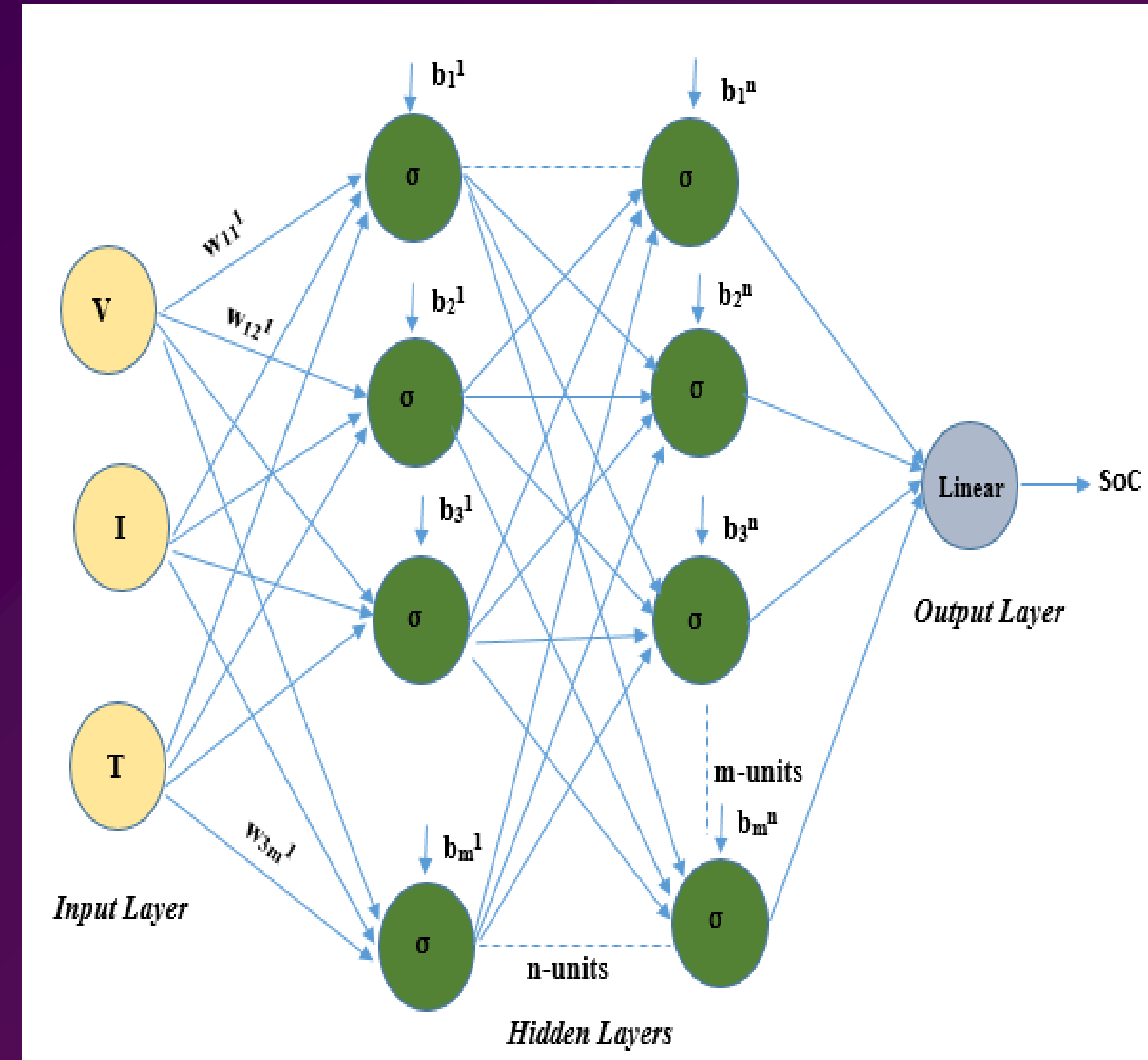
$$h_j = \sigma \left(\sum w_{ij} x_i + b_j \right)$$

σ denotes the ReLU activation function, expressed as:

$$\sigma(d) = \begin{cases} 0, & \text{if } d < 0 \\ d, & \text{if } d \geq 0 \end{cases}$$

SoC is calculated at the output layer as:

$$Y = \text{SoC} = \text{linear}(\sum w_{fg}^l h_g^{l-1} + b_g^l)$$





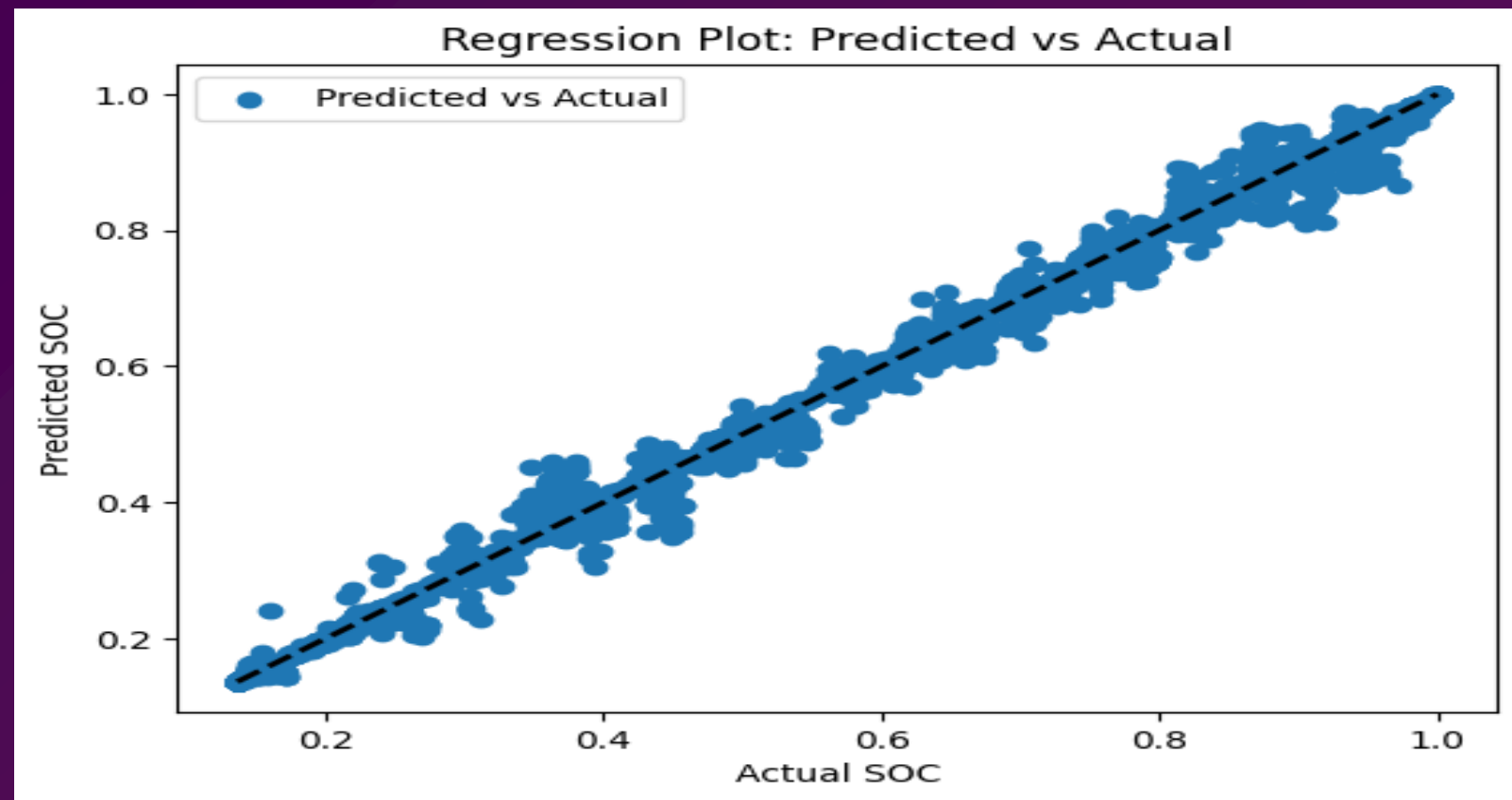
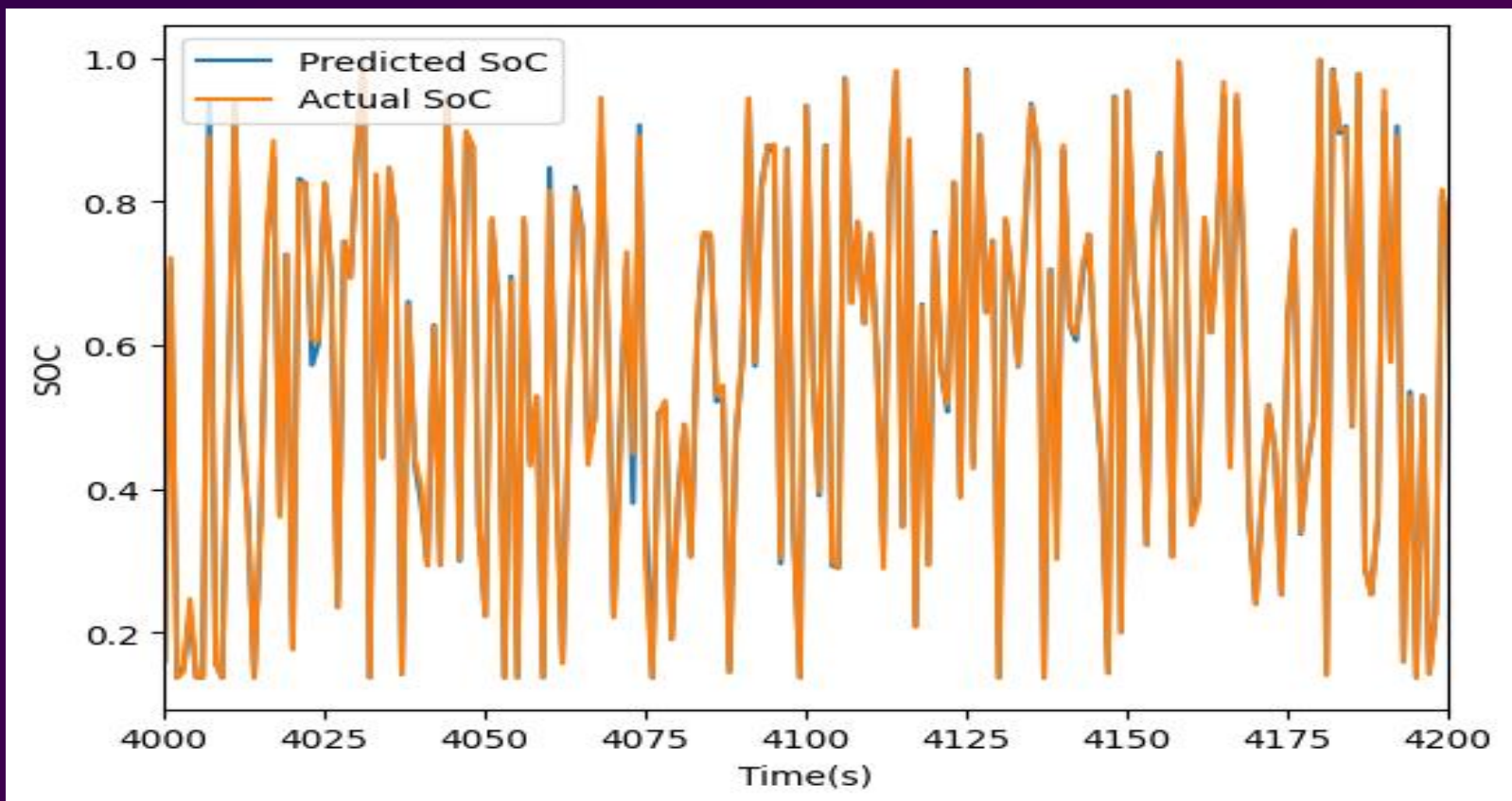
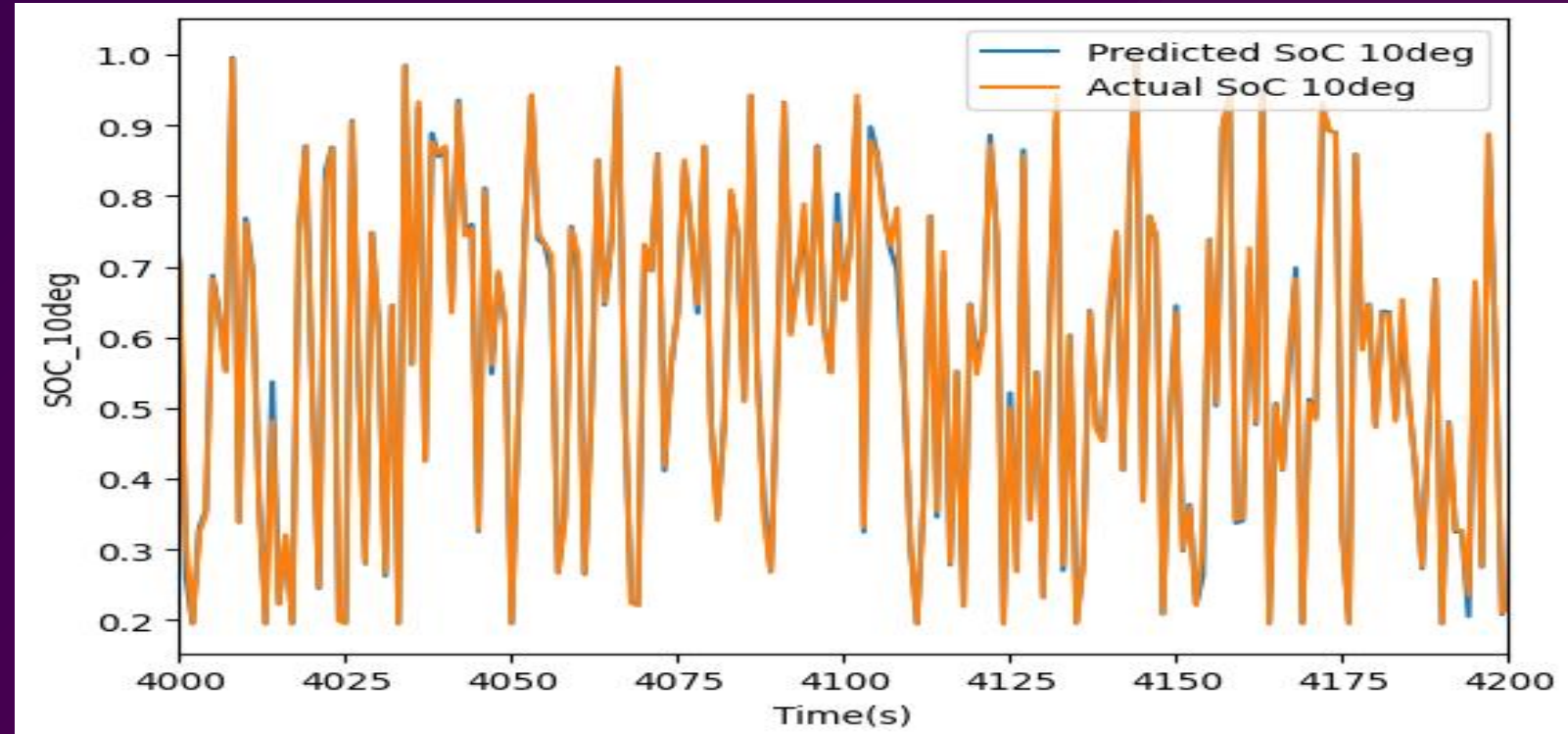
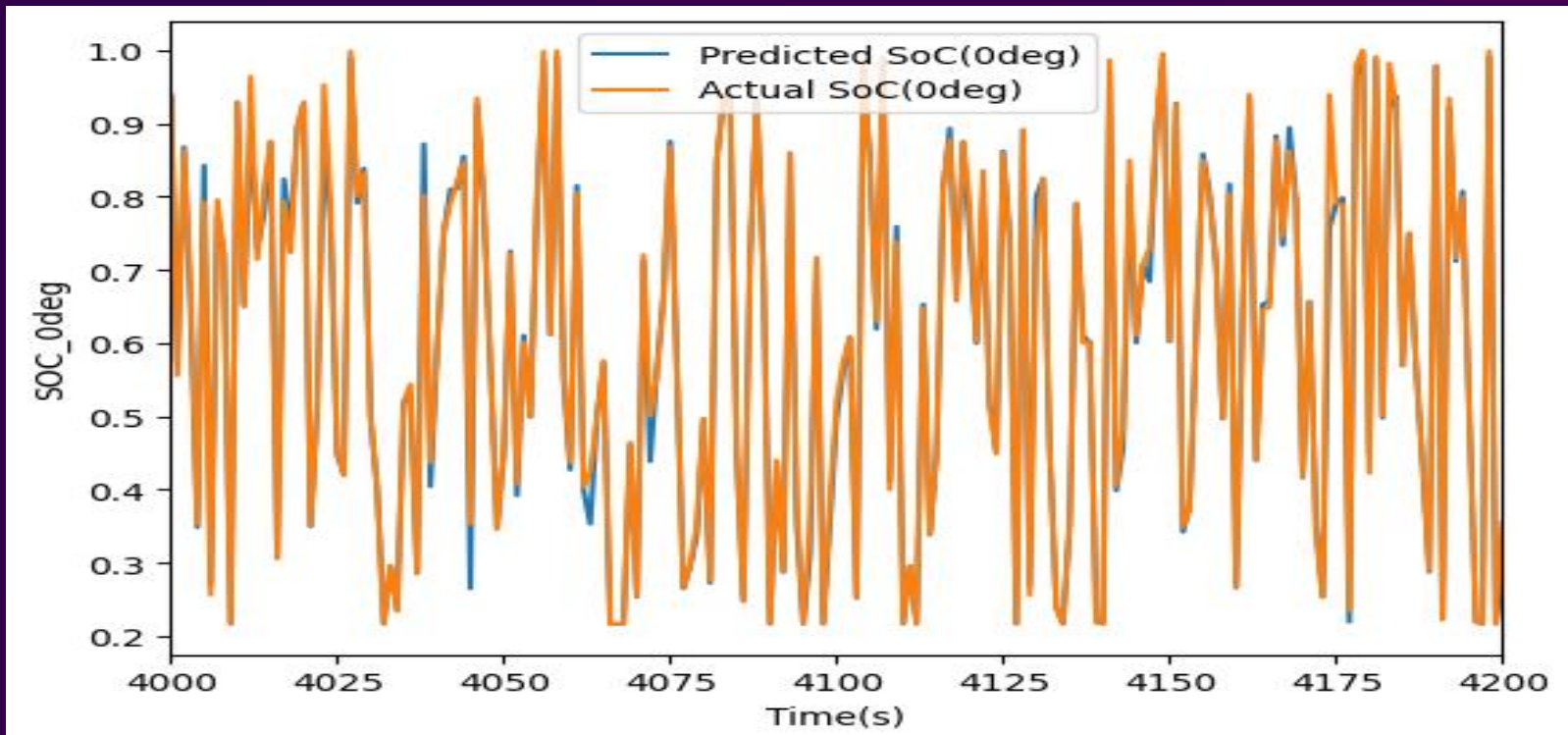
OPTIMAL HYPERPARAMETERS



Hyper-parameters	Values
Training algorithm	Backpropagation
Optimization algorithm	Adam
Number of hidden layers	2
Neurons per Hidden Layer	64
Activation Function	ReLU (Hidden Layer) Linear (Output Layer)
Batch Size	128
Cost Function	MAE



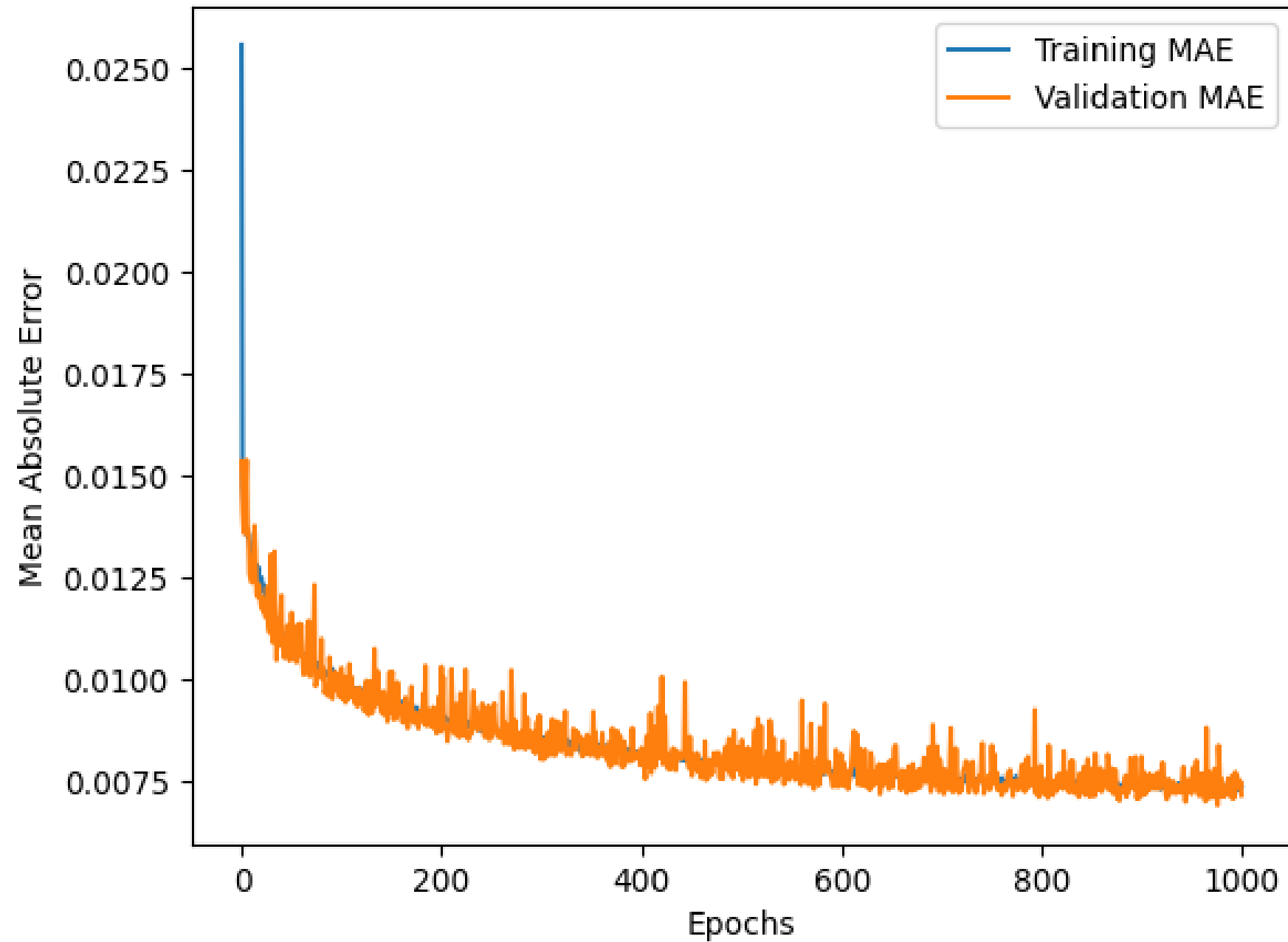
RESULTS & DISCUSSION



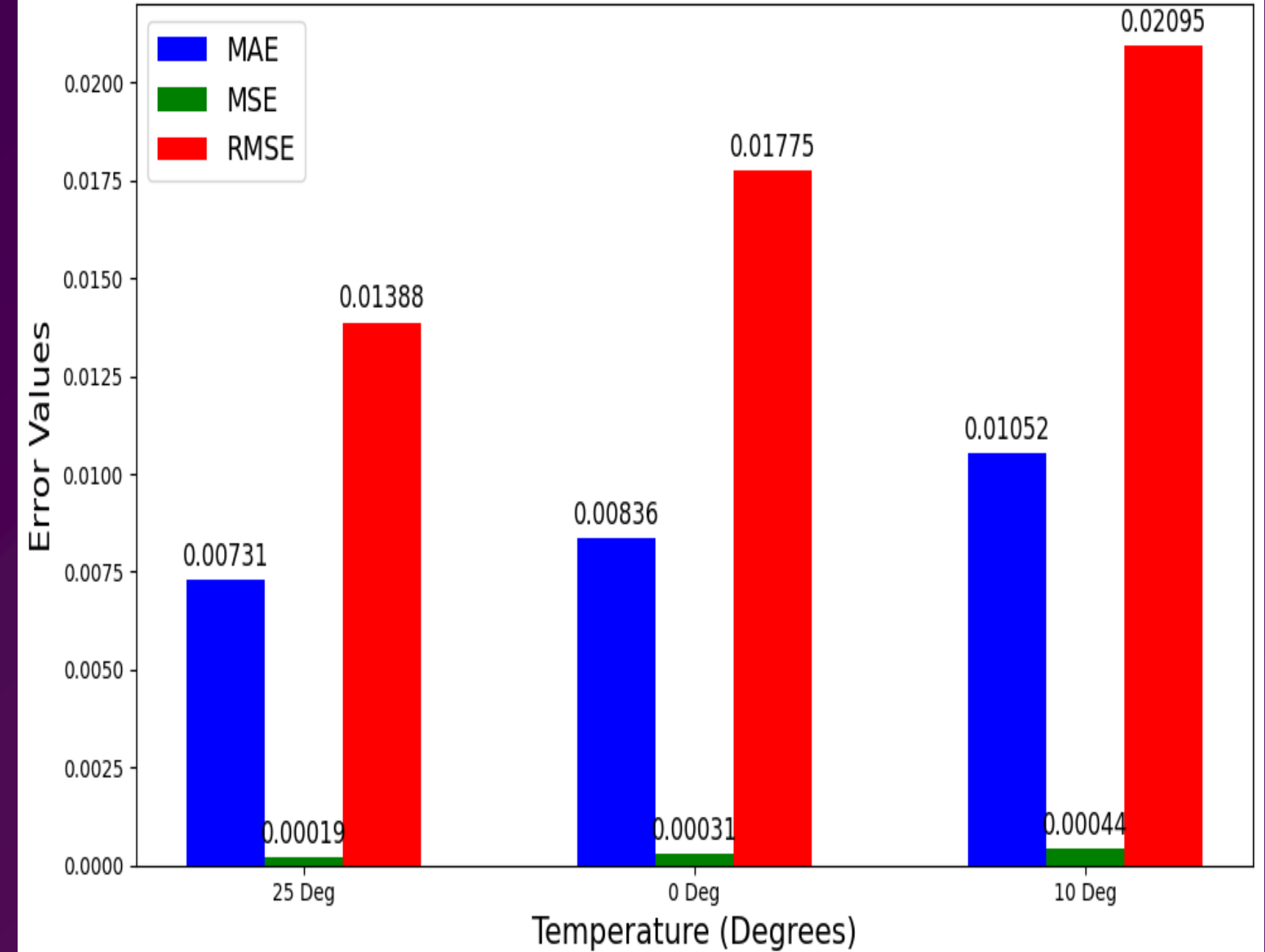


RESULTS & DISCUSSION

MAE Variation Over Epochs



MAE, MSE, and RMSE for Different Temperatures





RESULTS & DISCUSSION



Ref	Battery used	Model	Input parameters	Drive cycle	MAE (%)
[3]	PANASONIC 18650PF (NCA)	DNN	$V(t), I(t), T(t)$ $V_{avg}(t), I_{avg}(t)$	LA92, UDDS, HWFET, and US06	1.0567(0 ⁰ C) 1.0566 (10 ⁰ C) 1.0084 (25 ⁰ C)
[14]	PANASONIC 186650PF(NMC)	LSTM+ RNN	V_n, I_n, T_n	LA92, UDDS, HWFET, and US06	2.088(0 ⁰ C) 0.782 (10 ⁰ C) 0.774 (25 ⁰ C)
Implemented Approach	LG18650-HG2 (NMC)	DNN	V_x, I_x, T_x	LA92	0.083(0 ⁰ C) 0.105 (10 ⁰ C) 0.730 (25 ⁰ C)



CONCLUSION



- State of Charge (SOC) estimation is a critical component of Battery Management Systems (BMS), especially for lithium-ion batteries used in various applications.
- The study introduces a Deep Neural Network (DNN) model for SOC estimation, trained and validated using LA92 drive cycle data at three different temperatures: 0°C, 10°C, and 25°C.
- The results indicate that the model can precisely represent the non-linearity between SoC and measurable parameters with MAE of 0.830%, 1.050%, and 0.731% at 0°C, 10°C, and 25°C respectively.
- The proposed model demonstrates high accuracy in SOC estimation using parameters such as voltage, current, and temperature, outperforming existing methods in the literature.



FUTURE SCOPE



- Temporal information of the input data plays a vital role in a more accurate estimation of the SOC. Consequently, more sophisticated deep learning models such as LSTMS, BiLSTMS, and GRUS can be trained to leverage on the temporal information, which could potentially lead to better estimation.



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